

Lecture #19,20

Content to be covered

- ❖ Velocity Addition
- ❖ Relativity of Mass
 - ❑ Relation Between Mass and Energy
 - ❑ Relation Between Momentum and Energy
- ❖ Massless Particle

Resources to be consulted

- ❑ Concepts of Modern Physics by Arthur Beiser

Velocity Addition

Let us consider, something is moving along X-axis relative to both S (rest) and S' with U and U'. While S' frame is moving with velocity v. An observer in S measures three component of velocity to be

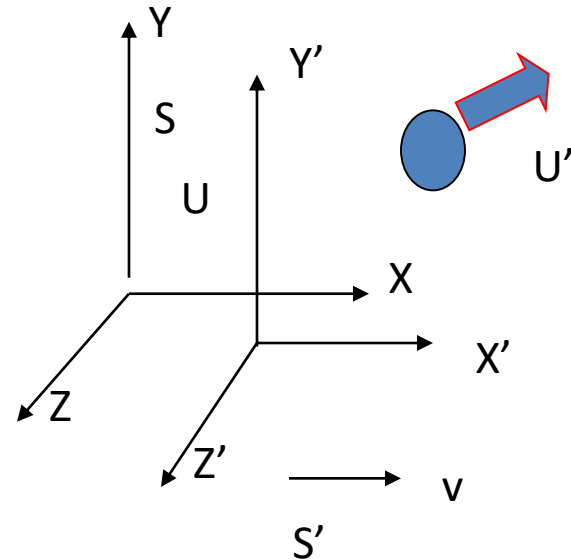
- To an observer in S' they are

$$U_x = \frac{dx}{dt}, \quad U_y = \frac{dy}{dt}, \quad U_z = \frac{dz}{dt}$$

- To an observer in S' they are

$$U'_x = \frac{dx'}{dt'}, \quad U'_y = \frac{dy'}{dt'}, \quad U'_z = \frac{dz'}{dt'}$$

$$x' = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}}, \quad y' = y, \quad z' = z, \quad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$



- By differentiating the Lorentz transform equations for x' , y' , z' and t' , we obtain

$$dx' = \frac{dx - vdt}{\sqrt{1 - \frac{v^2}{c^2}}}, \quad dy' = dy, \quad dz' = dz, \quad dt' = \frac{dt - \frac{vdx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

- Now we can write,

$$U'_x = \frac{dx'}{dt'} = \frac{dx - Vdt}{dt - \frac{Vdx}{c^2}} = \frac{\frac{dx}{dt} - V}{1 - \frac{V}{c^2} \frac{dx}{dt}} = \frac{U_x - V}{1 - \frac{VU_x}{c^2}}$$

- This is relativistic velocity transformation equation. Its Inverse transformation equation ($v \rightarrow -v$) is

$$U_x = \frac{U'_x + V}{1 + \frac{VU'_x}{c^2}}$$

- By applying the same technique we can obtain transformation for U_y and U_z as

$$U_y' = \frac{dy'}{dt'} = \frac{dy \sqrt{1 - \frac{V^2}{c^2}}}{1 - \frac{V dx}{c^2}} = \frac{\frac{dy}{dt} \sqrt{1 - \frac{V^2}{c^2}}}{1 - \frac{V}{c^2} \frac{dx}{dt}} = \frac{U_y \sqrt{1 - \frac{V^2}{c^2}}}{1 - \frac{V U_x}{c^2}}$$

Simiarly,

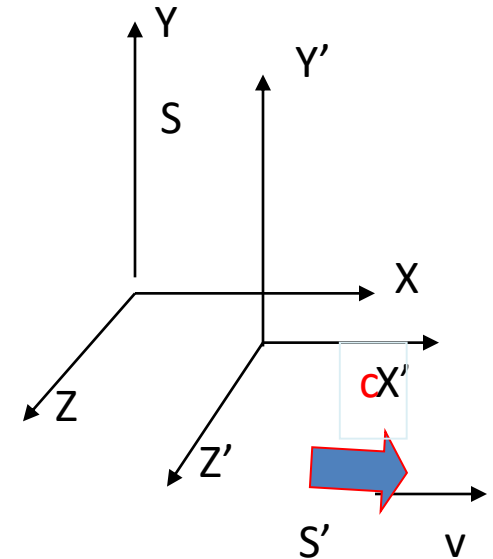
$$U_z' = \frac{U_z \sqrt{1 - \frac{V^2}{c^2}}}{1 - \frac{V U_x}{c^2}}$$

Example

- Let $U'_x = c$, means if a ray of light is emitted in the moving reference frame S' in the *opposite direction of motion relative to* S , an observer in frame S will measure the velocity

$$U_x = \frac{U'_x + V}{1 + \frac{VU'_x}{c^2}} = \frac{c + V}{1 + \frac{Vc}{c^2}} = \frac{c(c + V)}{(c + V)} = c$$

- Let $v = c$, again $U_x = c$.
- Let $U'_x = v = c$, again $U_x = c$.



Hence, it is impossible to exceed the speed of light beyond c and c is the upper limit of velocity of any particle.

Relativity of Mass

According to Newtonian mechanics the mass of a body is unaffected with change in velocity.

But space and time change.....

Therefore “mass” of a body is no longer be unaffected.

Relativity of Mass

Relativistic mass

$$m = \frac{m_o \longleftarrow \text{Mass at rest.}}{\sqrt{(1 - v^2 / c^2)}}$$

Rest mass is least

$$\text{Total Energy} = E = mc^2 = \frac{m_0 c^2}{\sqrt{1 - v^2 / c^2}}$$

$$\text{Rest mass energy or Potential energy} = m_0 c^2$$

A stationary body explodes into two fragments each of mass 1 kg that move apart at speeds of $0.6c$ relative to the original body. Find the mass of the original body.

Ans.: 2.5kg.

Relation Between Mass and Energy

Kinetic energy (T) of a moving body can be defined as the work done in bringing it from rest to its state of motion. That is:

$$T = \int_0^s F \cdot ds \quad \dots(1)$$

Where, F is the component of the applied force in the direction of the displacement ds and s is the distance over which the force acts. Using the relativistic form of the second law of motion.

$$F = \frac{d(mv)}{dt} \quad \dots(2)$$

Relation Between Mass and Energy

From eqn. (2)

$$F.ds = \frac{d(mv)}{dt} = \left(m \frac{dv}{dt} + v \frac{dm}{dt}\right).ds = \left(ma + v \frac{dm}{dt}\right).ds; \quad \text{where, } a = \frac{dv}{dt}$$

$$dE_k = m \frac{dv}{dt}.ds + v \frac{dm}{dt}.ds = m \frac{ds}{dt}.dv + v \frac{ds}{dt}.dm = mv dv + v^2 dm \dots\dots\dots(3)$$

we know that; $m = \frac{m_o}{\sqrt{1 - \frac{v^2}{c^2}}}$

On solving this eq., we have:

$$\left(1 - \frac{v^2}{c^2}\right)m = m_o^2 \Rightarrow c^2 m^2 - v^2 m^2 = m_o^2 c^2 \dots\dots\dots(4)$$

Differentiating eq.(4):

$$\begin{aligned} c^2 2m dm - 2vm^2 dv - 2mdmv^2 &= 0 \\ \Rightarrow 2m[dm(c^2 - v^2) - mv dv] &= 0 \\ \Rightarrow (c^2 - v^2)dm &= mv dv \dots\dots\dots(5) \end{aligned}$$

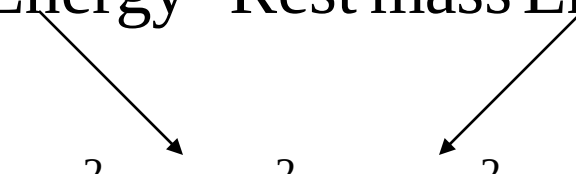
Substituting the value of $mv dv$ in eq.(3):

$$dE_k = (c^2 - v^2)dm + v^2 dm = c^2 dm \dots\dots\dots(6)$$

On integrating, we get $E_k = (m - m_o)c^2$

Relation Between Mass and Energy

Kinetic Energy = Total Energy - Rest mass Energy

$$K = \frac{m_0 c^2}{\sqrt{1 - v^2 / c^2}} - m_0 c^2 = mc^2 - m_0 c^2$$


This equation states that the kinetic energy of a body is equal to the increase in its mass, resulting from its relative motion, multiplied by the square of the speed of the light.

Therefore,

$$mc^2 = E_k + m_0 c^2$$

If we interpret mc^2 as the total energy E of the body, it follows that, when the body is at rest and $E_k=0$, it possesses the energy $E_0=m_0 c^2$ (rest energy or rest mass energy).

Limiting Case: When $v \ll c$

$$\text{Show that } K.E = \frac{1}{2} m_0 v^2$$

Limiting Case: When $v \rightarrow c$

K.E. tends to infinity. $\Rightarrow$ to accelerate the particle to the speed of light infinite amount of work would be needed to be done

Relation Between Momentum and Energy

Now we have

$$E = mc^2 = \frac{m_0 c^2}{\sqrt{1 - v^2 / c^2}} \Rightarrow E^2 = \frac{m_0^2 c^4}{(1 - v^2 / c^2)}$$

$$\text{and } p = mv = \frac{m_0 v}{\sqrt{1 - v^2 / c^2}} \Rightarrow p^2 c^2 = \frac{m_0^2 c^2 v^2}{(1 - v^2 / c^2)}$$

$$\Rightarrow E^2 - p^2 c^2 = \frac{m_0^2 c^4}{(1 - v^2 / c^2)} - \frac{m_0^2 v^2 c^2}{(1 - v^2 / c^2)}$$

$$E^2 - p^2 c^2 = \frac{m_0^2 c^4 - m_0^2 v^2 c^2}{(1 - v^2 / c^2)} = \frac{m_0^2 c^4 (1 - v^2 / c^2)}{(1 - v^2 / c^2)}$$

$$E^2 - p^2 c^2 = m_0^2 c^4$$

$$E = \sqrt{m_0^2 c^4 + p^2 c^2}$$

Relation Between Momentum and Energy

$$K = E - m_0 c^2$$

$$\text{But, } E = \sqrt{m_0^2 c^4 + p^2 c^2}$$

$$\therefore K = \sqrt{m_0^2 c^4 + p^2 c^2} - m_0 c^2$$

$$\text{or, } K = \left[\left(1 + \frac{p^2 c^2}{m_0^2 c^4} \right)^{1/2} - 1 \right] m_0 c^2$$

If $v < c$, that is, $p < m_0 c$, then

$$K \approx m_0 c^2 \left[\left(1 + \frac{p^2 c^2}{2 m_0^2 c^4} \right) - 1 \right]$$

Neglecting higher order terms of binomial expansion, we get

$$K = \frac{p^2}{2m_0} \quad \text{As } v \ll c, \text{ then } m_0 = m$$

$$K = \frac{p^2}{2m} \quad (\text{Classical Expression})$$



**Limiting value of
Relativistic Kinetic Energy**

Limiting Value of Relativistic Kinetic Energy

Relativistic kinetic energy is

$$K = mc^2 - m_0c^2$$
$$= \left[\left(1 - \frac{v^2}{c^2} \right)^{-1/2} - 1 \right] m_0c^2$$

Expanding by Binomial theorem

$$K = \left[1 + \frac{1}{2} \frac{v^2}{c^2} + \frac{3}{8} \frac{v^4}{c^4} + \dots - 1 \right] m_0c^2$$

If $v \ll c$ i.e. $v/c \ll 1$ So

$$K = \frac{1}{2} \frac{v^2}{c^2} m_0c^2 = \frac{1}{2} m_0v^2$$

This is expression for kinetic energy in non-relativistic case.

Massless Particle?

We have $E = mc^2 = \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}}$ and

$$p = mv = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}}$$

1) When $m_0=0$ and $v<c$, $E=0$, $p=0$

2) When $m_0=0$ and $v=c$, $E=0/0$, $p=0/0$ indeterminate form, hence must have E and p .

A particle which has zero rest mass

$$E = \sqrt{m_0^2 c^4 + p^2 c^2}$$

For massless particle , $m_0 = 0$

$$\therefore E = \sqrt{p^2 c^2} = pc \quad \text{or} \quad p = \frac{E}{c}$$

Since p is also equal to $mv = Ev / c^2$, therefore , $v = c$

i.e. the velocity of the massless particle is same as that of light in free space.

The energy of the particle , $E = pc = mc^2$

Where m is the mass equivalent to energy E , that is , $m = E / c^2$